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1.0 - Matrix Definitions

A **matrix** is a rectangular collection of numbers. For example:

$$A = \begin{pmatrix} 1 & -5 & 4 \\ 2 & 5 & 3 \end{pmatrix}$$

is a matrix with two rows and three columns. For that reason, it is called a 2 by 3 matrix. This is called the **dimension** of a matrix. It is expressed as number of rows × number of columns. So,

$$B = \begin{pmatrix} 1 & -5 & 4 \\ 5 & 3 & -8 \\ 1 & 5 & 4 \\ 2 & 5 & 3 \end{pmatrix}$$

is a 4 × 3 matrix. It is common to refer to elements in a matrix by subscripts, like so.

$$B = \begin{pmatrix} b_{1,1} & b_{1,2} & b_{1,3} \\ b_{2,1} & b_{2,2} & b_{2,3} \\ b_{3,1} & b_{3,2} & b_{3,3} \\ b_{4,1} & b_{4,2} & b_{4,3} \end{pmatrix}$$

With the row first and the column second. So in this case, $b_{2,1} = 5$ and $b_{1,3} = 4$. Generally, matrices are denoted as bold capital letters. A **vector** is a matrix with only one row (called a **row vector**)or only one column (called a **column vector**). For example:

$$C = \begin{pmatrix} 2 & 7 & -3 & 5 \end{pmatrix}$$

C is a 4 dimensional row vector.

$$D = \begin{pmatrix} 2 \\ 9 \\ -3 \\ 3 \\ 6 \end{pmatrix}$$

D is a 5 dimensional column vector. An "ordinary" number can be thought of as a 1 × 1 matrix, also known as a **scalar.** Some examples of scalars are shown below:

$$E = \pi$$

$$F = 6$$

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